

Krish Kumar

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SUMMARY

Computer Science and Engineering undergraduate and aspiring **DevOps Engineer / Software Development Engineer**, specializing in backend development and cloud-native systems. Experienced deploying microservices on **Kubernetes** using FastAPI and Flask. Built three end-to-end systems spanning self-service deployment tooling, distributed microservices, and relational database engineering, with a focus on CI/CD automation, container orchestration, and database design.

Core Competencies: Microservices Architecture, CI/CD Pipelines, Container Orchestration, Database Design.

EDUCATION

Thapar Institute of Engineering and Technology	08/2024 – 05/2028
Bachelor of Engineering in Computer Science and Engineering CGPA: 9.32/10.0	Patiala, Punjab, India
Swami Sant Dass Public School	04/2021 – 03/2023
Class XII, CBSE Score: 96.33%	Phagwara, Punjab, India

TECHNICAL SKILLS

Languages: Python, C, C++, SQL, JavaScript, Rust

Frameworks & Libraries: FastAPI, Flask, Flutter, SQLAlchemy, Jinja2, REST APIs, Microservices

DevOps & Cloud: Docker, Kubernetes, Helm, Jenkins, GitHub Actions, ArgoCD, GitOps, AWS EC2, NGINX Ingress, cert-manager, Prometheus, Grafana, Linux, Bash

Databases & Tools: PostgreSQL, MySQL, MongoDB, Redis, Git

Core CS Concepts: Data Structures & Algorithms, OOPs, Operating Systems, Computer Networks, DBMS

PROJECTS

CapstoneConnect | Microservices Capstone Group Finder 05/2026 – Present

FastAPI, Kubernetes, Redis, PostgreSQL, WebSockets

- Delivered a full-stack platform combining 3 core capabilities: profile discovery, automated group matching, and real-time chat, to help students form capstone project teams.
- Architected a **6-service FastAPI backend** on Kubernetes, routing traffic between independent, horizontally-scalable microservices through NGINX Ingress path rules.
- To prevent dropped chat messages across pod replicas, implemented a **Redis pub/sub backplane** that sustained **500+ concurrent** WebSocket connections locally.
- Because external API providers enforced strict rate limits, orchestrated a **CronJob-driven GitHub sync pipeline** with an in-memory caching layer to keep usage within bounds.
- Enforced transactional integrity for group-join requests in **PostgreSQL**, resolving race conditions across **100+ simulated concurrent requests**.

Nimbus | Self-Service Deployment Platform 03/2026 – Present

Python, FastAPI, Kubernetes, Docker, Kaniko

- Provisioned a self-service platform integrating FastAPI, Kubernetes, and Docker, letting developers deploy live applications without writing raw manifests.
- Automated complex deployment workflows by developing a **custom kopf controller** managing **3 CRDs** via a declarative reconciliation loop.
- Since deployed apps needed automatic TLS without manual setup, integrated **Kaniko-based in-cluster builds** with GitHub webhooks and **cert-manager**.
- Load-tested **Horizontal Pod Autoscaling** with k6 under a simulated 5x traffic spike, validating scale-out from **1 to 5 replicas**.

Hospital Management System | Role-Based Healthcare Platform 01/2026 – 04/2026

Flask, MySQL, Kubernetes, SQLAlchemy

- Shipped a role-based healthcare portal for tracking patient appointments, prescriptions, and administrative access.
- Designed an **8-table, BCNF-normalized MySQL schema** to support the system's core workflows.
- To enforce atomic, transactional appointment booking, wrote **7 stored procedures, 3 functions, and 6 triggers** in MySQL.
- Achieved **100% data integrity** across **20+ simulated pod restarts** by migrating relational database storage to a **Kubernetes StatefulSet** with persistent volumes.
- Automated schema setup with a Kubernetes Job, streamlining local environment provisioning.